From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013



ALL INDIA TEST SERIES

# CONCEPT RECAPITULATION TEST - IV

## Paper 1

**Time Allotted: 3 Hours** **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

**(i)** **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

**Section-A (11 to 15)** contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

**(ii)** **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | = | | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | = | | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | = | | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | = | | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

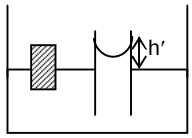
# Physics

PART – I

## SECTION – A
### Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In a capillary tube placed inside the liquid of density ($\rho$)in a container, the rise of liquid is h. When block of density '$\sigma$' is placed on the liquid as shown in figure, liquid in the tube is h′. If $\sigma < \rho$ then




(A) h′ = h (B) h′ < h
(C) h′ > h (D) insufficient data

2. Two points A and B are at distances of 'a' and 'b' respectively from an infinite conducting plate having charge density $\sigma$. The work done in moving charge Q from A to B is

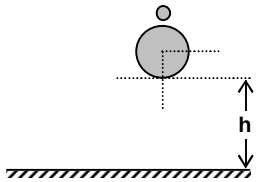
(A) $\frac{Q\sigma}{\varepsilon_0}(b-a)$ (B) $\frac{\sigma}{(b-a)}Q$

(C) $\frac{Q\sigma}{(b-a)\varepsilon_0}$ (D) none of these

3. A point charge of 0.1C is placed on the circumference of a non-conducting ring of radius 1m which is rotating with a constant angular acceleration of 1 rad/sec$^2$. If ring starts it's motion at t = 0 the magnetic field at the centre of the ring at t = 10 sec, is

(A) $10^{-6}$ T (B) $10^{-7}$ T
(C) $10^{-8}$ T (D) $10^{7}$ T

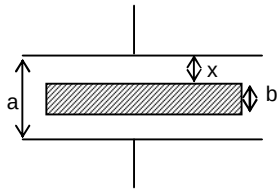
4. A small sphere and a big sphere are released from rest with a very small gap from height h as shown in the figure. The mass of bigger sphere is very large as compared to mass of smaller sphere. The height from the point of collision of smaller sphere with the bigger sphere to which the smaller sphere will rise if all the collisions are elastic

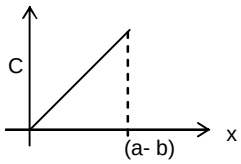


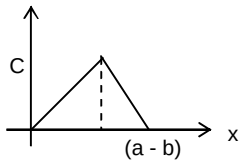

(A) 2h (B) 4h
(C) 6h (D) 9h

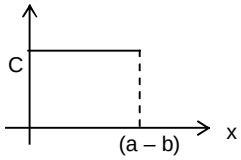
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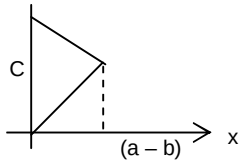
5. The distance between two parallel plates of a capacitor is a. A conductor of thickness b(b < a) is inserted between the plates as shown in the figure. The variation of effective capacitance between the surfaces of conductor and plate as a function of the distance (x) is best represented by




(A)


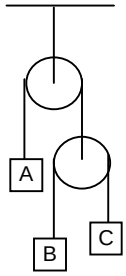

(B)



(C)



(D)



6. Three blocks A, B and C having masses m kg 2kg and 3 kg are attached by massless strings and ideal pulleys as shown in the figure. When the system is released from rest if the block 'A' remains stationary , the mass of block 'A' is

(A) 2.2 kg (B) 2.6 kg
(C) 2.4 kg (D) 4.8 kg

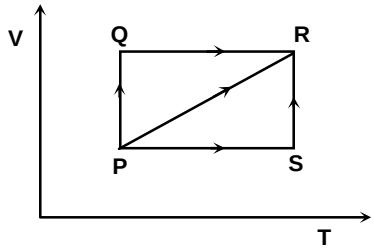



7. A block of mass 2 kg is attached to one end of a massless rod of length $\frac{1}{\pi}$m . The rod is fixed to a horizontal plane at the other end such that the block and rod are free to revolve on a horizontal plane. The coefficient of friction between the block and surface is 0.1. Block is made to rotate with uniform angular speed by applying a constant external force in tangential direction on the block. The work done by external force till the rod rotates by $90^0$ is

(A) 0 (B) 10 joule
(C) $\frac{\pi}{2}$ joule (D) 1 joule

***Space for Rough work***

8. One mole of a monoatomic gas is taken from P to R, via three paths PQR, PR and PSR. If work done by the gas in PQR is $W_1$, in PR work done is $W_2$ and in PSR work done is $W_3$, then

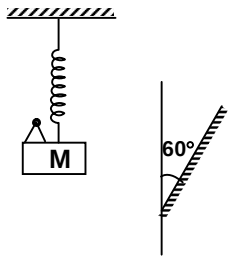



(A) $W_2 > W_3 > W_1$
(B) $W_1 > W_2 > W_3$
(C) $W_2 > W_1 > W_3$
(D) $W_3 > W_2 > W_1$

9. A block of wood of relative density 0.5 is placed 10 m inside a vessel containing water. The vessel is accelerated upward with an acceleration of 10 $m/s^2$. If the block is released at some instant, then the time taken by block to reach the surface of water from that instant is (g = 10 $m/s^2$)

(A) 0.5 sec
(B) 1 sec
(C) 2 sec
(D) none of these

10. An insect of negligible mass is sitting on a block of mass M, tied with a spring of force constant k. The block performs simple harmonic motion with amplitude A infront of a plane mirror placed as shown in the figure. The maximum speed of insect relative to its image will be




(A) $A\sqrt{\frac{k}{m}}$
(B) $\frac{A\sqrt{3}}{2}\sqrt{\frac{k}{m}}$
(C) $A\sqrt{3}\sqrt{\frac{k}{m}}$
(D) $A\sqrt{\frac{m}{k}}$

***Space for Rough work***

## Multiple Correct Answer(s) Type

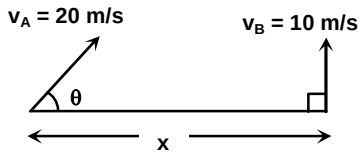
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. When an electron of hydrogen like atom jumps from a higher energy level to a lower energy level. Then,
(A) angular momentum of the electron remains constant.
(B) kinetic energy increases.
(C) wavelength of de-Broglie wave, associated with motion of the electron, decreases.
(D) all of the above

12. If the filament current in Coolidge tube is increased by increasing the voltage in the filament circuit then
(A) $\lambda_{min}$ remain same
(B) Max. K.E. of electron not affected.
(C) Intensity increases
(D) Intensity decreases

13. A parallel plate capacitor of plate area 0.2 $m^2$ and spacing $10^{-2}$ m is charged to $10^3$ volt and then disconnected from the battery, and pulled apart to double the plate spacing
(A) work required to pull the plate is $8.8 \times 10^{-5}$ J
(B) final voltage on the capacitor is $2 \times 10^3$ Volts.
(C) Final charge on the capacitor becomes two times of initial
(D) final charge becomes half of initial value.

14. Two particles A and B are projected simultaneously in the direction shown in the figure with velocities $v_A$ = 20 m/s and $v_B$ = 10 m/s respectively. They collide in air after 1/2 sec. Then




(A) the angle $\theta$ is 60°
(B) the distance x is $5\sqrt{3}$ m
(C) the angle $\theta$ is 30°
(D) The distance x is $15\sqrt{3}$ m

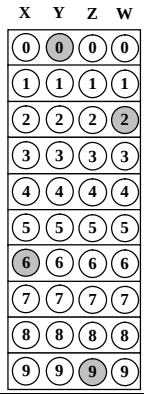
15. A uniform square plate of mass m and edge a initially at rest starts rotating about one of the edge under the action of a constant torque $\tau$. Then at the end of the 5th sec after start
(A) angular momentum is equal to $5\tau$
(B) kinetic energy is equal to $\frac{75\tau^2}{ma^2}$
(C) angular momentum is equal to $2.5\tau$
(D) kinetic energy is equal to $\frac{75\tau^2}{2ma^2}$

***Space for Rough work***

## SECTION –C

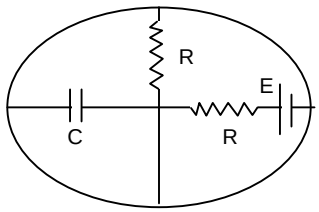
### Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.




1. A light rod of length L, is hanging from the vertical smooth wall of a vehicle moving with acceleration $\sqrt{3}$ g having a small mass attached at it's one end is free to rotate about an axis passing through the other end. The minimum velocity given to the mass at it's equilibrium position so that it can complete vertical circular motion is $\sqrt{KgL}$ . Find the value of K.

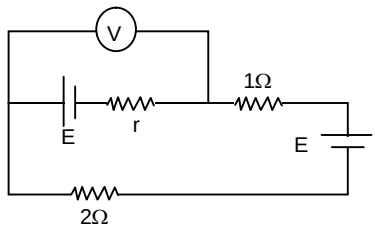
2. Find energy stored in the capacitor in it's steady state.




3. The potential difference between two points A and B which are separated by a distance of 1m along the field in a uniform electric field of $10NC^{-1}$ is 5K Volt. Find the value of K.

4. The power factor of a circuit in which a box having unknown electrical devices connected in series with a resistor of resistance $3\Omega$ is 3/5. Find the reactance of the box in $\Omega$.

5. In the given circuit the reading of ideal voltmeter is E/2. Find the internal resistance of the battery in $\Omega$.




***Space for Rough work***

# Chemistry

PART – II

## SECTION – A

### Straight Objective Type

This section contains **10 multiple choice** questions numbered **1 to 10.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

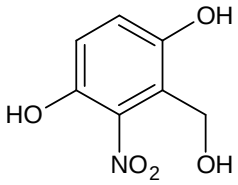
1. $K_c$ for the reaction $[Ag(CN)_2]^{-1} \rightleftharpoons Ag^+ + 2CN^-$, the equilibrium constant at $25^oC$ is $4 \times 10^{-19}$, then the silver ion concentration in the solution which was originally 0.1 M in KCN and 0.03 M in $AgNO_3$ is
   (A) $7.5 \times 10^{18}$ (B) $7.5 \times 10^{-18}$
   (C) $7.5 \times 10^{19}$ (D) $7.5 \times 10^{-19}$

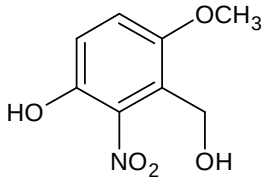
2. A gas in an open container is heated from $27^oC$ to $127^oC$. The fraction of the original amount of gas left in the container would be:
   (A) 3/4 (B) 1/2
   (C) 2/3 (D) 7/8

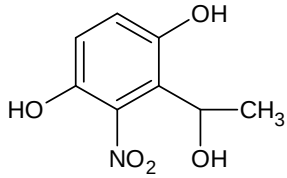
3. Consider the equilibrium, $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g) + \text{heat}$
   If $O_2$ is added and volume of the reaction vessel is increased, the equilibrium will
   (A) shift in forward direction (B) shift in backward direction
   (C) remain unchanged (D) be unpredictable

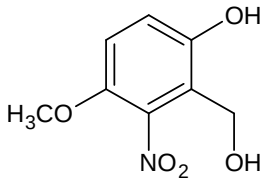
4. If the energy of H-atom in the ground state is – E, the velocity of photo electron emitted when a photon having energy $E_p$ striked a stationary $Li^{2+}$ ion in ground state, is given by:
   (A) $V = \sqrt{\frac{2(E_p - E)}{m}}$ (B) $V = \sqrt{\frac{2(E_p + 9E)}{m}}$
   (C) $V = \sqrt{\frac{2(E_p - 9E)}{m}}$ (D) $V = \sqrt{\frac{2(E_p - 3E)}{m}}$

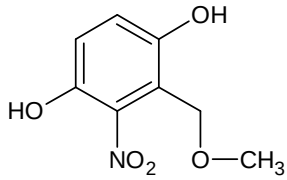
**Rough Work**

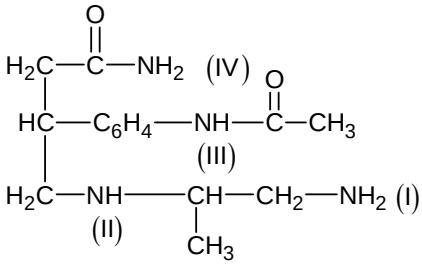
5.  $\xrightarrow[(1\,eqv.)]{CH_2N_2}$ A, A is :

(A) 

(B) 

(C) 

(D) 

6. 

Arrange the above mentioned nitrogen atoms of the compound in decreasing order of basic nature?

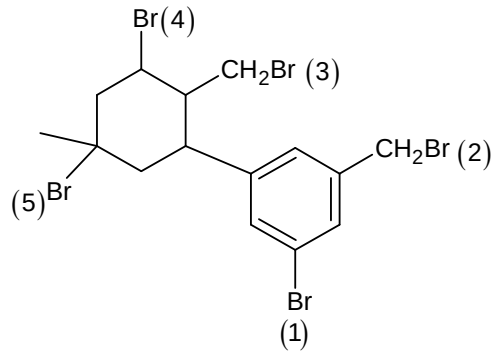
(A) I > II > IV > III
(B) II > I > IV > III
(C) II > I > III > IV
(D) I > IV > III > II

7. Which of the following compounds can form silicones on hydrolysis?

(A) $(CH_3)_2SiCl_2$
(B) $(SiH_3)_3N$
(C) $SiCl_4$
(D) All of the above

**Rough Work**

8.




Arrange the above mentioned bromine atoms in decreasing order of their reactivity towards hydrolysis through $S_N1$ path.

(A) 2 > 1 > 5 > 4 > 3 (B) 5 > 2 > 1 > 3 > 4

(C) 2 > 5 > 4 > 3 > 1 (D) 5 > 4 > 2 > 1 > 3

9. Which of the following compounds on decomposition, releases two gases containing the same elements?

(A) $FeSO_4$ (B) $FeCO_3$

(C) $Fe(NO_3)_2$ (D) $Fe(ClO_3)_2$

10. Gold number gives

(A) amount of gold in colloid.

(B) weight in mg of colloid required to break gold sol.

(C) weight in mg of colloid to protect gold sol.

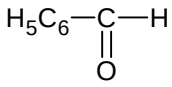
(D) amount of NaCl required to coagulate gold sol.

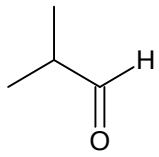
**Rough Work**

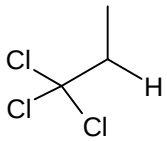
**Multiple Correct Choice Type**

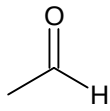
This section contains **5 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. Which of the following compounds does not disproportionate on treating with conc. NaOH?

(A) 

(B) 

(C) 

(D) 

12. If electrolysis of 1 M $NiSO_4$ is carried with inert electrodes at pH = 7 at 25°C (the pressure of gases are taken 1 bar).

$\left(E^{o}_{Ni^{2+}/Ni} = -0.25\ V \qquad E^{o}_{H_2O/H_2} = -0.83\ v\right)$

The product of electrolysis at cathode and at anode (initially) are:

(A) Ni at cathode
(B) $H_2$ at cathode
(C) $O_2$ at anode
(D) $S_2O_8^{2-}$ at anode

13. Which of the following is/are correct:

(A) $[CrCl_2(CN)_2(NH_3)_2]$ and $[CrCl_3(NH_3)_3]$, both have $d^2sp^3$ hybridization.
(B) Magnetic moment of $[PdCl_4]^{2-}$ is zero.
(C) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$, all have $dsp^2$ hybridization of central metal.
(D) It is difficult to substitute chelating ligands compared to similar monodentate ligands.

14. Which of the following is/are correct?

(A) On strong heating $FeCl_3$ gives $FeCl_2$ and $Cl_2$.
(B) Dilute $CuCl_2$ is blue due to formation of $[CuCl_4]^{2-}$.
(C) $PH_3$ can be obtained by heating red P with NaOH.
(D) $PH_3$, $H_3PO_3$ and $H_3PO_2$, all are reducing agents.

15. $2X(g) + Y(g) + 3Z(g) \longrightarrow$ Products

The rate equation of the above reaction is given by:

Rate = $K[X][Y]^o[Z]^2$

Choose the correct statements:

(A) The rate of reaction becomes doubled by doubling the conc. of (X).
(B) Rate of reaction decreases by reducing the conc. of (Y) to half of the original value.
(C) The half life of (Z) increases by increasing its conc.
(D) If 75% of (X) undergoes reaction in 20 sec., 50% of (X) will react in 10 sec.

**Rough Work**

## SECTION – C
## Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

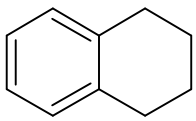
| X | Y | Z | W |
|---|---|---|---|
| 0 | **0** | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | **2** |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| **6** | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | **9** | 9 |

1. A compound (A) having molecular formula $C_7H_{11}Br$ is optically active. A reacts with HBr in the absence of peroxide to give isomeric products (B) and (C). Treating (A) with potassium-t-butoxide gives (D). (D) on reductive ozonolysis gives two moles of formaldehyde and one mole of 1, 3 cyclopentane dione. (A) in the presence of peroxide reacts with HBr to give (E). The total number of chiral carbon in one molecule of (E) is:

2. How many of the following nitrates give positive brown ring test as well as oxygen on heating?
$KNO_3$, $Pb(NO_3)_2$, $NH_4NO_3$, $Ca(NO_3)_2$, $Sr(NO_3)_2$, $Mg(NO_3)_2$, $NaNO_3$

3. How many (S = O) are present in $H_2S_2O_5$?

4. A complex of potassium, iron and cyanide ions $K_x[Fe(CN)_6]$ is 100% ionized at 1 molal. If its elevation in boiling points is 2.08°C, find out the value of 'x' in the complex.
(Given $K_b = 0.52^oC\ mol^{-1}\ kg$)

5. 
$\xrightarrow[\text{(excess)}]{\text{NBS}}$ X $\xrightarrow[\text{excess}]{\text{KOH/H}_2\text{O}}$ (Y) (unstable) $\xrightarrow{\Delta}$ Z

The $sp^2$-hybridized carbon atoms present in one molecule of the end product Z is:

**Rough Work**

# Mathematics

PART – III

## SECTION – A

### Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The value of $\int_{-2}^{1}\left[x\left[1+\cos\left(\frac{\pi x}{2}\right)\right]+1\right]dx$, (where [.] denotes the greatest integer function), is

(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) none of these

2. $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x)=\begin{bmatrix}\min\{f(t):0\le t\le x\} & ; & 0\le x\le 1\\ 3-x & ; & 1<x\le 2\end{bmatrix}$ then $g\left(\frac{1}{4}\right)+g\left(\frac{3}{4}\right)+g\left(\frac{5}{4}\right)$ has the value equal to

(A) $\frac{7}{4}$ (B) $\frac{9}{4}$

(C) $\frac{13}{4}$ (D) $\frac{5}{2}$

3. If $x_1, x_2, x_3, \ldots, x_n$ are in A.P. whose common difference is $\alpha$, then the value of $\sin\alpha(\sec x_1 \sec x_2 + \sec x_2 \sec x_3 + \ldots + \sec x_{n-1} \sec x_n)$ is

(A) $\frac{\sin(n-1)\alpha}{\cos x_1 \cos x_n}$ (B) $\frac{\sin n\alpha}{\cos x_1 \cos x_n}$

(C) $\sin(n-1)\alpha \cos x_1 \cos x_n$ (D) $\sin n\alpha \cos x_1 \cos x_n$

***Space for rough work***

4. In $\Delta ABC$, the median AD divides $\angle BAC$ such that $\angle BAD : \angle CAD = 2 : 1$. Then $\cos\left(\frac{A}{3}\right)$ is equal to

(A) $\frac{\sin B}{2\sin C}$ (B) $\frac{\sin C}{2\sin B}$

(C) $\frac{2\sin B}{\sin C}$ (D) none of these

5. The projection of the line $\frac{x+1}{-1} = \frac{y}{2} = \frac{z-1}{3}$ on the plane $x - 2y + z = 6$ is the line of intersection of this plane with the plane

(A) $2x + y + z = 0$ (B) $3x + y - z = 2$

(C) $2x - 3y + 8z = 3$ (D) none of these

6. If $|z_2 + iz_1| = |z_1| + |z_2|$, $|z_1| = 5$ and $|z_2| = 12$, then area of $\Delta ABC$, if affixes of A, B and C are $z_1$, $z_2$ and $\left[\frac{(z_2 - iz_1)}{(1-i)}\right]$ respectively, is

(A) $\frac{13}{2}$ (B) 0

(C) $\frac{169}{2}$ (D) $\frac{169}{4}$

7. A person buys eight packets of TIDE detergent. Each packet contains one coupon, which bears one of the letters of the word TIDE. If he shows all the letters of the word TIDE, he gets one free packet. If he gets exactly one free packet, then the number of different possible combinations of the coupons is

(A) ${}^7C_3 - 1$ (B) ${}^8C_4 - 1$

(C) ${}^8C_3$ (D) $4^4$

8. If the chord of contact of tangents from a point P to a given circle passes through Q, then the circle on PQ as diameter

(A) cuts the given circle orthogonally (B) touches the given circle externally

(C) touches the given circle internally (D) none of these

***Space for rough work***

9. An ellipse is sliding along the co–ordinate axes. If at an instant, the foci of the ellipse are (1, 1) and (3, 3), then area of the director circle of the ellipse (in sq. units) is
(A) $2\pi$ (B) $4\pi$
(C) $6\pi$ (D) $8\pi$

10. The coefficient of the term independent of x in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is
(A) 210 (B) 105
(C) 70 (D) 112

**Multiple Correct Choice Type**

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

11. $P(A)=\frac{3}{8}$; $P(B)=\frac{1}{2}$; $P(A\cup B)=\frac{5}{8}$, which of the following do/does hold good?
(A) $P\left(\frac{A^C}{B}\right)=2P\left(\frac{A}{B^C}\right)$ (B) $P(B)=P\left(\frac{A}{B}\right)$
(C) $15P\left(\frac{A^C}{B^C}\right)=8P\left(\frac{B}{A^C}\right)$ (D) $P\left(\frac{A}{B^C}\right)=P(A\cap B)$

12. Given that x + y + z = 15 when a, x, y, z, b are in A.P. and $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{5}{3}$ when a, x, y, z, b are in H.P. then
(A) G.M. of a and b is 3 (B) one possible value of a + 2b is 11
(C) A.M. of a and b is 6 (D) greatest value of a – b is 8

13. A quadrilateral is inscribed in parabola, then
(A) quadrilateral may be cyclic
(B) diagonals of the quadrilateral may be equation
(C) all possible pairs of adjacent sides may be perpendicular
(D) none of these

14. $f(x) = x^2 - 2ax + a(a + 1)$; $f : [a, \infty) \to [a, \infty)$. If one of the solutions of the equation $f(x) = f^{-1}(x)$ is 5049, then the other may be
(A) 5051 (B) 5048
(C) 5052 (D) 5050

***Space for rough work***

15. If $\int \frac{e^{x-1}}{(x^2-5x+4)} 2xdx$ = A.F(x – 1) + B.F(x – 4) + C and $F(x) = \int \frac{e^x}{x} dx$, then

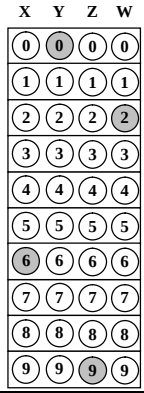
(A) $A = -\frac{2}{3}$ (B) $B = \left(\frac{4}{3}\right)e^3$

(C) $A = \frac{2}{3}$ (D) $B = \left(\frac{8}{3}\right)e^3$

## SECTION – C

### Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.




1. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Also curve passes through the point (1, 1). Then the length of intercept of the curve on the x–axis is _____

2. At the point P(a, $a^n$) on the graph of $y = x^n$ ($n \in N$) in the first quadrant a normal is drawn. The normal intersects the y–axis at the point (0, b). If $\lim_{a\to 0} b = \frac{1}{2}$, then n is equal to _____

3. $f : R \to R$, $f(x) = \frac{3x^2+mx+n}{x^2+1}$. If the range of this function is [–4, 3), then the value of |m + n| is _____

4. If $a_1, a_2, a_3, \ldots\ldots, a_{12}$ are in A.P. and $\Delta_1 = \begin{vmatrix} a_1a_5 & a_1 & a_2 \\ a_2a_6 & a_2 & a_3 \\ a_3a_7 & a_3 & a_4 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_2a_{10} & a_2 & a_3 \\ a_3a_{11} & a_3 & a_4 \\ a_3a_{12} & a_4 & a_5 \end{vmatrix}$ then $\Delta_1 : \Delta_2$ is equal to _____

5. Number of solution of the equation $\sin^4 x - \cos^2 x \sin x + 2\sin^2 x + \sin x = 0$ in $0 \le x \le 3\pi$ is _____

***Space for rough work***